Pinned group summary: `SO(n=5, q=2)`

Pinning-Sympy automated output

July 6, 2026

Contents

1 Basic attributes

Group	`SO(n=5, q=2)`
Matrix size	5×5
Rank	2

1.1 Symmetric bilinear form

Dimension	5
Witt index	2
Primitive element	N/A
Epsilon	N/A

Matrix: $\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & c_0 \end{bmatrix}$

1.2 Generic Lie algebra element

$$\begin{bmatrix} x_{00} & x_{01} & 0 & -x_{12} & -c_0x_{42} \\ x_{10} & x_{11} & x_{12} & 0 & -c_0x_{43} \\ 0 & -x_{30} & -x_{00} & -x_{10} & -c_0x_{40} \\ x_{30} & 0 & -x_{01} & -x_{11} & -c_0x_{41} \\ x_{40} & x_{41} & x_{42} & x_{43} & 0 \end{bmatrix}$$

1.3 Generic torus element

$$\begin{bmatrix} t_0 & 0 & 0 & 0 & 0 \\ 0 & t_1 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{t_0} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{t_1} & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

1.4 Trivial characters

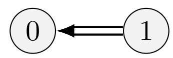
$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	`B2`
Reduced	True
Simply laced	False
Number of roots	8

2.1 Dynkin diagram



2.2 Cartan matrix

$$\begin{bmatrix} 2 & -1 \\ -2 & 2 \end{bmatrix}$$

2.3 Roots and coroots

Root	**Coroot**	**Simple**	**Sign**	**Multipliable**
$(-1,\ -1,\ 0,\ 0,\ 0)$	$(-1,\ -1,\ 1,\ 1,\ 0)$	No	$-$	No
$(-1,\ 0,\ 0,\ 0,\ 0)$	$(-2,\ 0,\ 2,\ 0,\ 0)$	No	$-$	No
$(-1,\ 1,\ 0,\ 0,\ 0)$	$(-1,\ 1,\ 1,\ -1,\ 0)$	No	$-$	No
$(0,\ -1,\ 0,\ 0,\ 0)$	$(0,\ -2,\ 0,\ 2,\ 0)$	No	$-$	No
$(0,\ 1,\ 0,\ 0,\ 0)$	$(0,\ 2,\ 0,\ -2,\ 0)$	Yes	$+$	No
$(1,\ -1,\ 0,\ 0,\ 0)$	$(1,\ -1,\ -1,\ 1,\ 0)$	Yes	$+$	No
$(1,\ 0,\ 0,\ 0,\ 0)$	$(2,\ 0,\ -2,\ 0,\ 0)$	No	$+$	No
$(1,\ 1,\ 0,\ 0,\ 0)$	$(1,\ 1,\ -1,\ -1,\ 0)$	No	$+$	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ, the set of positive integral linear combinations of α, β that are also roots is denoted

$$(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$$

α	β	Pairs (i, j)
$(-1,\ -1,\ 0,\ 0,\ 0)$	$(0,\ 1,\ 0,\ 0,\ 0)$	$(1,1),(1,2)$
$(-1,\ -1,\ 0,\ 0,\ 0)$	$(1,\ 0,\ 0,\ 0,\ 0)$	$(1,1),(1,2)$
$(-1,\ 0,\ 0,\ 0,\ 0)$	$(0,\ -1,\ 0,\ 0,\ 0)$	$(1,1)$
$(-1,\ 0,\ 0,\ 0,\ 0)$	$(0,\ 1,\ 0,\ 0,\ 0)$	$(1,1)$
$(-1,\ 0,\ 0,\ 0,\ 0)$	$(1,\ -1,\ 0,\ 0,\ 0)$	$(1,1),(2,1)$
$(-1,\ 0,\ 0,\ 0,\ 0)$	$(1,\ 1,\ 0,\ 0,\ 0)$	$(1,1),(2,1)$
$(-1,\ 1,\ 0,\ 0,\ 0)$	$(0,\ -1,\ 0,\ 0,\ 0)$	$(1,1),(1,2)$
$(-1,\ 1,\ 0,\ 0,\ 0)$	$(1,\ 0,\ 0,\ 0,\ 0)$	$(1,1),(1,2)$
$(0,\ -1,\ 0,\ 0,\ 0)$	$(1,\ 0,\ 0,\ 0,\ 0)$	$(1,1)$
$(0,\ -1,\ 0,\ 0,\ 0)$	$(1,\ 1,\ 0,\ 0,\ 0)$	$(1,1),(2,1)$
$(0,\ 1,\ 0,\ 0,\ 0)$	$(1,\ -1,\ 0,\ 0,\ 0)$	$(1,1),(2,1)$
$(0,\ 1,\ 0,\ 0,\ 0)$	$(1,\ 0,\ 0,\ 0,\ 0)$	$(1,1)$